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22IS642

JSS MAHAVIDYAPEETHA
JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

VI Semester BE Model Question Paper
Department of Information Science and Engineering
PROGRAMMING WITH C#

Duration: 3 hours

Max. Marks: 100

NOTE: Part A is Compulsory. Part B has internal choice.

PART – A

Q.No	CO	CD	PI	QUESTION	MARKS
1.	CO1	Understand	1.7.1	Describe the Four types of Punctuators in C# with suitable examples.	10
2a.	CO2	Understand	1.7.1	Explain Generics with an example.	04
2b.	CO2	Apply	2.5.2	Develop a generic method to exchange the positions of two different elements in an array.	06
3.	CO3	Understand	1.7.1	Explain Overloading of Logical operators with suitable examples.	10
4.	CO4	Understand	1.7.1	Explain the two mandatory steps involved in creating and displaying the main window in a program and Illustrate the functionality of the Application class with an example.	10
5.	CO5	Understand	1.7.1	Explain the Core Concepts and Architecture of ADO.NET and Discuss its benefits.	10

PART – B

Q.No	CO	CD	PI	QUESTION	MARKS
6a.	CO1	Understand	1.7.1	Explain the different Class Modifiers with examples.	04
6b.	CO1	Apply	2.5.2	Declare a class 'Student' with data members 'Name' and 'USN'. Declare another class 'Details' which inherits class 'Student'. Display Name and USN with Address and Email-id by creating the object of class 'Details'. Develop a C# program to implement this.	06
OR					

7a.	CO1	Understand	1.7.1	Explain any Four Member Modifiers with examples.	04
7b.	CO1	Apply	2.5.2	Create a class 'Rectangle' that contains 'Length' and 'Breadth' as data members. From this class, derive a class named 'Box' that has additional data member 'Depth'. The members of the class 'Rectangle' should consist of a constructor and a method Area() and 'Box' should have a constructor. Develop a C# program to implement this.	06
8a.	CO2	Understand	1.7.1	Demonstrate a Two Dimensional Array with a code snippet.	06
8b.	CO2	Apply	2.5.2	Write a C# program to display the digits of an integer in words.	04
OR					
9a.	CO2	Apply	1.7.1	Demonstrate a Two Dimensional Jagged Array with a code snippet.	04
9b.	CO2	Understand	2.5.2	Explain the three common types of Collections provided by the .NET Framework.	06
10.	CO3	Apply	2.5.2	Illustrate a Delegate with three methods performing string modification such as Replacing spaces with hyphens, Removing spaces and Reversing a string.	10
OR					
11.	CO3	Understand	1.7.1	Explain a Structured Exception Handling with a code snippet.	10
12.	CO4	Understand	2.5.2	Develop a new Windows Form that is capable of handling the following events: a. Respond to the MouseMove and MouseDown events, b. Capture and process keyboard input via the KeyUp event.	10
OR					
13.	CO4	Understand	1.7.1	Write the steps involved to develop a Windows Forms application as shown below using Visual Studio (next page).	10

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14.	CO5	Apply	2.5.2	Create a Company database and explain the database connectivity to C# program to display Employee details.	10
OR					
15.	CO5	Apply	2.5.2	Create a Employee database and write SQL statements for the following: i) Retrieve all EmpID and Emp-name columns. ii) Retrieve all Emp--name whose Salary > 5LPA. iii) Insert a new record into the Employee table. iv) Display all the records.	10

Course Outcome (CO):	
CO-1	Demonstrate C# language features, Types, Inheritance and Polymorphism.
CO-2	Develop applications using multi-dimensional arrays such as Jagged arrays, Collections and Generics.
CO-3	Apply concepts of Operator Overloading, Delegates, Events and Exception Handling.
CO-4	Develop GUI based applications for Mobile and Desktop.
CO-5	Create distributed data-driven applications using the .NET Framework, C#, SQL Server and ADO.NET

Performance Indicator (PI):	
1.7.1	Apply theory and principles of computer science and engineering to solve an engineering problem.
2.5.2	Identify processes/modules/algorithms of a computer-based system and parameters to solve a problem.

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